

Design and Implementation of PID Controller Based Ball and Beam System

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Abstract: This project is about the design and implementation of a ball and beam system using a Proportional-Integral-Derivative (PID) controller. The system is intended to keep a ball steady in a position on a beam, and automatically adjusts the angle to do so. An ultrasonic sensor obtains ball position and an input signal to an arduino uno is used to direct the processing of the sensor data and the calculation of an ultrasonic sensor output controlling the PID controller. Then a servo motor adjusts the beam angle to position the ball to the desired position. The PID parameters are considered for stability of position tracking with low overshoot and steady state error. A prototype of the developed controller is then tested at various setpoints to assess control performance. This project illustrates what is required for using PID control in a real-time closed loop system.

Keywords: PID control, ball and beam.

I. INTRODUCTION

Control systems keep a system at a desired condition by continually measuring its output and acting to correct it. A closed loop control system is such that the output is compared to the set point and the error is minimized by using feedback. The ball and beam system is a real-life application of a closed-loop control system. It is made with a ball on a beam which is adjustable from a servo motor applied to the angle of the beam with respect to the horizontal. If the beam has an inclination the ball will slide along the beam if gravity occurs. So the angle of the beam can be used to manipulate the position of the ball. The instability of this system is the essential problem of this system. When the ball is not controlled, it drifts away from where it's supposed to be. The necessary control action is therefore determined by a PID controller. The proportional, integral and derivative terms

improve the response, minimize the steady state error and eliminate unwanted oscillations. In this project, the position of the ball is measured by an ultrasonic sensor which then transmits this measurement to an Arduino Uno microcontroller. The Arduino reads the actual position and subtracts it from the setpoint in order to produce the PID output. This output drives a servo motor to adjust the beam angle and move the ball towards the desired position. A similar setup with the arduino and servo motor and ultrasonic sensor is provided in the supporting study. This project aims to construct and experiment with a working prototype of a ball and beam system and compare the effectiveness of position control using a PID controller. The system's performance is judged by its ability to follow the desired position, reduce overshoot, reduce settling time and to keep steady state error down.

II. DESIGN SPECIFICATIONS, SYSTEM REQUIREMENTS AND CONSTRAINTS OF THE SYSTEM

A. Design Specifications

The primary objective of the system is to regulate the position of a ball along a beam at a user-defined reference (setpoint) using closed-loop PID control, and to reject external disturbances (e.g., a manual push on the ball) so that the ball returns to the setpoint automatically. The following functional specifications were defined for the design:

- **Reference tracking:** The system shall maintain the ball at a desired setpoint distance from the sensor, with the nominal operating setpoint chosen at the midpoint of the beam (15 cm).
- **Disturbance rejection:** The controller shall be able to return the ball to the setpoint within a reasonable settling time after an external disturbance is applied.
- **Real-time feedback control:** The ball's position shall be measured

continuously and fed back to the Arduino microcontroller, which computes the PID control signal in real time.

- **Actuation:** The beam's tilt angle shall be adjusted by a servo motor connected directly at one end of the beam (a single-point lever/tilt mechanism), based on the PID output.
- **Acceptable tolerance:** Given the sensing method used (see constraints below), the system is designed to keep the ball within a practical steady-state error band around the setpoint, rather than achieving zero steady-state error.

B. System Requirements

Mechanical Requirements

- A rigid support base to hold the beam and servo assembly steady during operation.
- A beam/track long enough to give the ball adequate travel range for meaningful control demonstration, with low enough friction to allow smooth rolling.
- A ball light enough to respond quickly to beam tilt, enabling visible dynamic response.
- A servo-driven tilt mechanism capable of changing the beam angle proportionally to the PID controller's output.

Electrical/Electronic Requirements

- A microcontroller (Arduino Uno) capable of reading sensor input, executing the PID algorithm, and generating a PWM control signal for the servo motor.
- A distance sensor capable of continuously measuring the ball's position and reporting it to the Arduino in a usable form.

- A servo motor accepting PWM input and providing sufficient torque to tilt the beam under the ball's weight.
- A stable power supply for the Arduino and servo motor.

Control/Software Requirements

- Implementation of the PID control algorithm on the Arduino, computing the control action from the error between the measured ball position and the setpoint.
- Sufficiently fast sensor sampling and control loop update rate to track the ball's motion without noticeable lag.
- Provision to change the setpoint (reference position) in software.

C. System Constraints

1. Sensor Selection Trade-off (Cost vs. Accuracy)

An ultrasonic distance sensor was selected over an infrared (IR) sensor primarily due to its lower cost and easier availability. This is a deliberate design trade-off: ultrasonic sensors are more economical but are inherently less robust to environmental variation compared to IR sensors.

2. Environmental Sensitivity of the Ultrasonic Sensor

Since the ultrasonic sensor measures distance based on the time-of-flight of a sound wave, its accuracy depends on the speed of sound in air, which varies with air temperature, humidity, and air density. Because the prototype operates in an open, uncontrolled environment (not a sealed or climate-controlled lab enclosure), air disturbances and density variation introduce measurement noise into the sensor readings.

3. Observed Steady-State Error As a direct consequence of the above constraint, with the setpoint fixed at 15 cm, the system exhibits a steady-state positional error of approximately ± 2 cm around the setpoint under open-air conditions. This is a sensing-limitation-driven error rather than a controller tuning deficiency

— i.e., it originates from noise/variation in the measured input to the PID loop, not from incorrect $K_p/K_i/K_d$ values.

4. Mechanical/Material Constraints

- The beam/track is constructed from a lightweight plastic material rather than metal, chosen for ease of fabrication and low cost. This keeps the moving mass low but makes the beam more susceptible to minor flex and surface irregularities affecting ball rolling behavior.
- The base structure is wooden, providing adequate rigidity for a benchtop prototype but limited long-term durability compared to a metal chassis.
- The ball itself is a lightweight plastic ball, which responds quickly to beam tilt but is also more easily perturbed by air currents and minor surface imperfections on the track, compounding the effect of sensor noise.

5. Cost Constraint Component selection throughout the project (sensor, servo, microcontroller, base and beam materials) was guided by the need to build a functional, low-cost educational prototype rather than an industrial-grade precision system, which directly explains several of the accuracy trade-offs described above.

III. SYSTEM PRACTICAL MODEL

The practical model is divided into two primary components: the mechanical assembly and the electrical system. The mechanical section comprises the beam, lever arm, and DC servomotor horn, while the electrical section incorporates the DC servomotor, an ultrasonic sensor, and an Arduino microcontroller.

3.1 Mechanical Subsystem

Selecting appropriate materials is a foundational step in any engineering project. As illustrated in Figure 3, the primary mechanical components consist of the base support, beam, and ball.

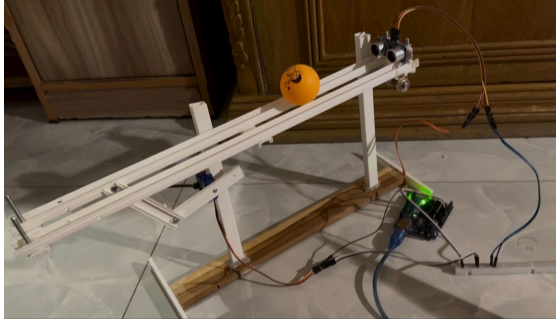


Figure 3: Ball and Beam System Practical Model.

The physical system is constructed using the following materials and specifications:

Wooden Base Support: 14 inch (35 cm) in length

PVC Beam: 43 cm in length by 5 cm in width.

Servo Motor Horn

Lever Horn

The detailed dimensions of the complete physical model are depicted in Figure 4.

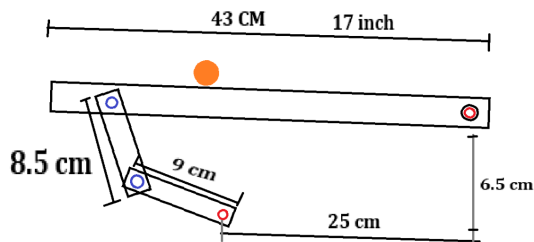


Figure 4: Dimensions of the physical model

3.2 Electrical Subsystem

The electrical system comprises the following primary components:

1. Arduino Uno Microcontroller
2. DC Servo Motor (SG 90)
3. Ultrasonic Sensor

The ultrasonic sensor is connected to digital pin 7. It transmits its digital signal to the Arduino, which divides the signal duration by 58 to determine the ball's distance in centimeters. This calculated position serves as

the feedback input for the PID controller, where it is compared against the target setpoint.

layout-wise, the DC servo motor is connected to pin 9, leveraging its Pulse Width Modulation (PWM) capability. It receives its commanded angle directly from the PID controller output. The complete circuit layout and electrical connections are illustrated in Figure 5.

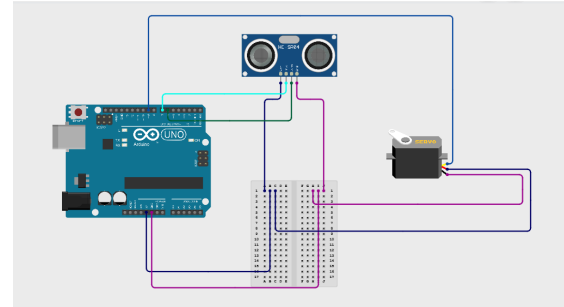


Figure 5 : Electrical Connection for the Ball and Beam System

IV. IDENTIFICATION OF ALTERNATIVE APPROACHES AND SELECTION OF THE OPTIMAL SOLUTION

A. Distance Measurement Sensors Alternatives:

- a. Infrared Distance Sensor: Measures position using light reflection angle. Infrared (IR) sensors are used to detect and measure the presence of an object within a given area.
- b. Ultrasonic Sensor (selected): Measures time-of-flight using sound waves. The sensor calculates distance based on the time taken for the sound waves to return. Ultrasonic sensors can measure distances with high accuracy.

A. Control Methodology Alternatives:

- a. Open Loop Control: The servo motor receives a fixed command under open-loop control without first

determining the position of the ball. If the ball is moved, the system would not correct its position automatically. Therefore, open-loop control is not suitable for this project.

- b. Proportional (P) Control : It responds directly to the current distance error. It applies small or large levels of correction based on the distance between the ball and the set-point. Using only proportional control may result in a small position error or unwanted oscillation.
- c. Proportional-Integral (PI) Control: It uses both the current error and the past errors. The integral term continuously corrects the small errors by adding incremental tilt to force small errors to zero. This causes the system to overshoot even more violently than Proportional control.
- d. Proportional-Derivative (PD) Control: It combines current error correction and rate-of-change sensing. The derivative term calculates the ball's speed toward or away from the set-point and generates an opposite tilt angle. In this case, tiny mechanical friction gaps in the beam can leave minor steady state positional error.
- e. Proportional-Integral-Derivative (PID) Control (selected): PID control combines three types of control action. Proportional responds to the current position error, Integral clears out the past errors over time, and Derivative senses the rate of change and provides a certain stop based on current speed.

C. The PID control is selected because-

- i. It requires low computational power.
- ii. It provides good position control.
- iii. It can reduce steady-state error.
- iv. It can improve system stability.

V. PID Controller Design

A. Role of the Three PID Components

A Proportional-Integral-Derivative (PID) controller acts as the brain of the Ball and Beam system. Its main task is to observe the ball's position, compare it to the desired set-point, and continuously command the servo motor to tilt the beam so that the ball stays balanced at the exact target location.

The controller repeatedly reads the sensor data each second, calculates the distance error, and combines three corrective actions to maintain the ball's stability.

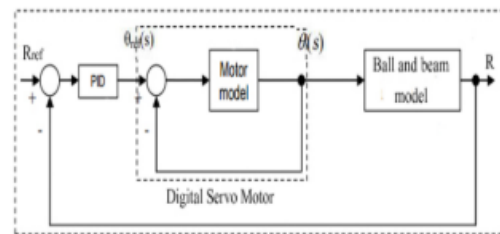


Figure: Overall system

The proportional action measures the ball's distance from the target point and tilt the beam sharply or mildly based on the distance. The integral action focuses on the error that build over time. If the ball remains slightly off target due to any physical faults, it gradually adds extra tilts until the ball reaches the exact target point. The derivative action determines the ball's speed. If the ball is moving quickly toward the target, it tilts the beam in the opposite direction early on, acting as a physical brake.

B. PID tuning

PID tuning is an important part of the project.

1. First, set the integral and derivative gains to zero, gradually increase the proportional gain and observe how quickly the ball moves toward the desired position.
2. If the ball oscillates around the desired position, gradually increase the derivative gain.

3. If a small position error remains after the ball becomes stable, gradually increase the integral gain.
4. Test the controller at different set points.
5. Move the ball away from its desired position and observe whether the controller returns it to the target point.

Experiment no.	Controller Parameter		
	kp	ki	kd
1	1	0	0
2	2	0	0
3	3	0	0.1
4	1	0	0.2
5	2	2	2
6	3	0	1
7	1	2	0.2
8	2	0	0.1
9	3	1	0
10	5	2	2
11	5	1	0

VI. Prototype Development and Implementation

The prototype was developed to make a practical ball and beam control system. The main idea is to keep the ball at a particular position on the beam by using a PID controller. The system is a closed loop system as the position of the ball is recorded and re-recorded and compared to the desired position.

A. Prototype Development

There are two major parts of the prototype – the mechanical part and the electrical part. The mechanical section consists of the base support, beam, ball, lever and servo motor horn. The electrical components are primarily an Arduino Uno, a DC servo motor and an ultrasonic sensor.

A wooden base support of 14 inch (35 cm) long was used in making the mechanical structure. This was accomplished using a PVC beam 43 cm long and 5 cm wide. The ball may be rolled on the beam. Also a servo motor horn and a lever horn are used to attach the motor to the beam. As the servo motor spins, the angle of the beam will change and the ball will slide along the beam due to the force of gravity.

B. Electrical Implementation

The Arduino Uno is used as the main controller in the electrical part. The ultrasonic sensor is used to detect the distance between the ball and the sensor. It is plugged at digital pin #7 of the Arduino. The output of the PID is then taken as the feedback for the PID controller.

The angle of the beam is changed using a DC servo motor (SG90). It is connected to the Arduino's pin 9 which has PWM capability. The difference between desired position and actual position is used by the PID to determine the movement of the servo motor.

C. Working Principle

The process of the prototype is very simple. The desired position or set point is first provided. After that, the ultrasonic sensor is able to detect the exact position of the ball. The Arduino then calculates this position and the desired position and determines the error.

The error may be defined as follows:

$$[e(t)=r(t)-y(t)]$$

The desired position in question is where ($r(t)$), and the actual position is ($y(t)$).

This error is used by the PID controller to determine the action to take. The servo motor then adjusts the angle of the beam, causing the ball to move towards the desired position. This process continues as the ball remains very close to the set point.

D. PID Controller Implementation

PID controller is composed of three sections: proportional, integral, and derivative control. These three parts are used for the enhancement of position control of the ball.

The PID parameters were optimized and tested in the experiment through the use of the prototype. For the recorded test, the controller parameters were:

$$K_p = 2.000$$

$$K_i = 0.000$$

$$K_d = 0.100$$

E. MATLAB Monitoring

MATLAB was also used to see and plot the real-time response of the system. The Arduino was linked to the computer to be able to keep track of the location of the ball. In this way, it would be possible to compare the different responses for the different PID values, and the more optimal controller values could be chosen.

When tested at different set points of 10 cm, 15 cm, and 20 cm, small oscillations and no overshoot were noticed at 10 cm. The overshoot and settling time were greater at 20 cm than at 10 cm.

Overall, the prototype combines the mechanical and electrical parts to make a working ball and beam control system. The Arduino controls the position of the ball with the help of the ultrasonic sensor, servo motor and PID controller. Various PID values were experimented to achieve a good response from the system. The system was also validated by testing with various locations and the results demonstrated that the ball position could be controlled properly with the prototype system.

VII. Experimental Results & Analysis

A. Experimental Setup and Response Measurement

The experiment tested a ball-and-beam system controlled by a PID controller implemented on an Arduino Uno microcontroller. The objective of the experiment was to move a ball along a beam to a specified reference position and keep it there. An ultrasonic distance sensor measured the real-time position of the ball along the beam and provided feedback to the controller. The PID controller processed the position error and generated a PWM signal to drive a servo motor, which adjusted the beam angle accordingly.

Two separate experimental runs were recorded. In both runs, the reference setpoint was approximately 15 cm from the sensor. The controller gains used were $K_p = 2.000$, $K_i = 0.000$, and $K_d = 0.100$ for both tests. The performance of the system was evaluated based on the following parameters: reference position, final position, steady-state error, maximum absolute error, RMSE, peak position, peak time, overshoot, rise time, and settling time.

B. PID System Response

The PID controller was tuned with the following gains for both experimental runs: $K_p = 2.000$, $K_i = 0.000$, and $K_d = 0.100$. The proportional gain was set to generate a corrective response proportional to the position error. No integral action was used ($K_i = 0$), meaning the steady-state offset caused by friction or gravity was not automatically corrected. The derivative gain ($K_d = 0.100$) was included to provide damping and reduce oscillation speed.

Response 1 is shown in Fig. 1. The reference position was 15.000 cm. The ball reached a final position of 18.025 cm, leaving a steady-state error of 3.025 cm. The maximum absolute error recorded during the response was 23.913 cm. The RMSE for this response was 4.557 cm. The ball overshoot the reference significantly, reaching a peak position of

38.913 cm at a peak time of 29.469 s. The overshoot was 159.42%. The rise time was 0.942 s. The settling time was not obtained in the recorded response.

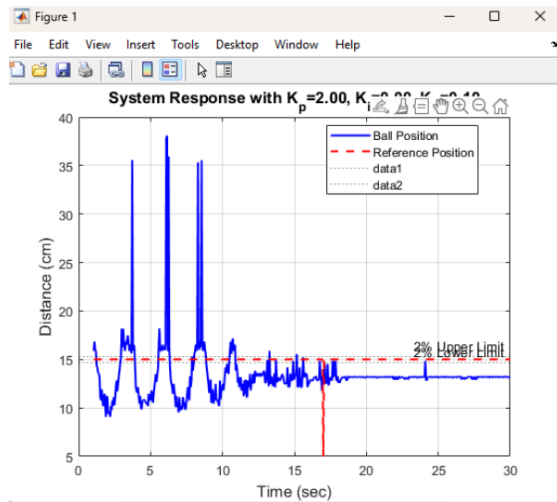


Fig. 1. PID system response for the first experimental condition ($K_p = 2.000$, $K_i = 0.000$, $K_d = 0.100$; Reference = 15.000 cm).

Response 2 is shown in Fig. 2. The reference position was 14.983 cm. The ball reached a final position of 13.176 cm, resulting in a steady-state error of 1.807 cm. The maximum absolute error was 23.107 cm. The RMSE for this response was 3.216 cm. The peak position was 38.090 cm, reached at a peak time of 6.147 s. The overshoot was 154.22%. The rise time was 0.000 s. As with Response 1, the settling time was not obtained in the recorded response.

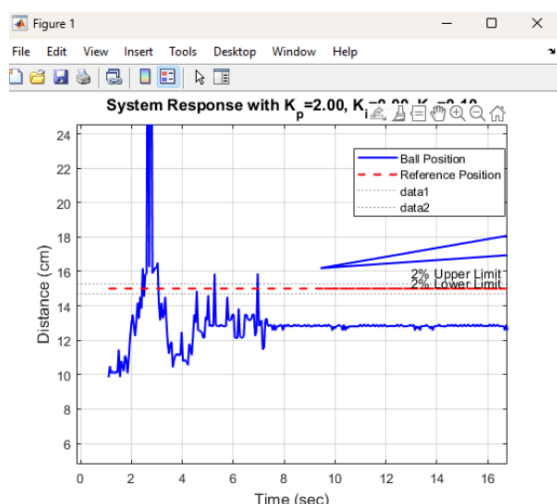


Fig. 2. PID system response for the second experimental condition ($K_p = 2.000$, $K_i = 0.000$, $K_d = 0.100$; Reference = 14.983 cm).

TABLE I PID RESPONSE PERFORMANCE SUMMARY

Parameter	Response 1	Response 2	Unit
K_p	2.000	2.000	—
K_i	0.000	0.000	—
K_d	0.100	0.100	—
Reference Position	15.000	14.983	cm
Final Position	18.025	13.176	cm
Steady-State Error	3.025	1.807	cm
Max. Absolute Error	23.913	23.107	cm
RMSE	4.557	3.216	cm
Peak Position	38.913	38.090	cm
Peak Time	29.469	6.147	s
Overshoot	159.42	154.22	%
Rise Time	0.942	0.000	s
Settling Time	NaN (Not Met)	NaN (Not Met)	s

C. Performance Analysis

In both experimental runs, the ball was driven toward the reference position, confirming that the PID controller was able to produce a corrective response. However, the system did not achieve stable, settled control within the recording period, which is reflected in the key performance parameters.

The steady-state error for Response 1 was 3.025 cm, meaning the ball came to rest 3.025 cm away from the 15.000 cm reference. Response 2 had a smaller steady-state error of 1.807 cm, showing slightly better final positioning. Since $K_i = 0$ in both tests, there was no integral action to automatically correct for the remaining offset. This is a key reason why a steady-state error persisted in both cases.

The overshoot was very high in both responses: 159.42% in Response 1 and

154.22% in Response 2. This means the ball moved far past the reference position before turning back. In Response 1, the ball reached a peak position of 38.913 cm against a reference of 15.000 cm — more than double the target distance. In Response 2, the peak position was 38.090 cm. Such large overshoot values indicate that the proportional gain caused a strong initial drive but the derivative gain ($K_d = 0.100$) was not large enough to adequately slow the ball as it approached the setpoint.

The RMSE (Root Mean Square Error) was 4.557 cm for Response 1 and 3.216 cm for Response 2. RMSE measures the overall deviation of the ball position from the reference throughout the entire response. A higher RMSE indicates larger tracking error over time. The lower RMSE in Response 2 suggests that, on average, the ball stayed closer to the reference position in that run.

The maximum absolute error was 23.913 cm in Response 1 and 23.107 cm in Response 2. These values confirm that the ball moved very far from the reference at some point during each run, consistent with the high overshoot measured.

The rise time for Response 1 was 0.942 s, which shows the system responded relatively quickly to the step input. The rise time for Response 2 was recorded as 0.000 s, which may indicate that the ball was already near or crossing the reference level at the very start of that recording. In both cases, the fast rise confirms that the proportional action was effective in driving the ball toward the target quickly.

The settling time was not obtained in either response. This means the ball did not remain within an acceptable error band around the reference for a sustained period during the recording window. The absence of integral action and the large oscillation caused by high overshoot prevented the system from settling. The performance summary categorizes the overshoot as HIGH ($> 6\%$), the steady-state error as HIGH, and the settling time as NOT

MET — confirming that the current gain values are insufficient for stable, precise control.

D. Comparison of Responses

Table I provides a direct numerical comparison of both responses. Both runs used identical PID gains ($K_p = 2.000$, $K_i = 0.000$, $K_d = 0.100$), so the differences in results reflect the effect of slightly different initial conditions or disturbances during each run.

Response 2 showed a lower steady-state error (1.807 cm) compared to Response 1 (3.025 cm), indicating that the ball came to rest closer to the target in the second run. Response 2 also had a lower RMSE (3.216 cm vs. 4.557 cm), meaning the overall tracking accuracy was better throughout the second run.

In terms of overshoot, Response 2 (154.22%) performed marginally better than Response 1 (159.42%), though both values are very high. The peak positions were also similar: 38.090 cm for Response 2 and 38.913 cm for Response 1, indicating that in both cases the ball overshoot to nearly the same extent.

The most notable difference was the peak time: Response 1 reached its peak at 29.469 s, while Response 2 reached its peak much earlier at 6.147 s. This difference suggests that the initial dynamics of the two runs were different, possibly due to the starting position of the ball or slight variations in the physical setup at the start of each test.

Neither response achieved a satisfactory settling time. Both responses share the same fundamental limitation: the absence of integral control ($K_i = 0$) means that friction and gravity-related biases are not automatically removed, and the low derivative gain is insufficient to fully dampen the oscillations generated by the proportional term.

E. Overall Observation

The experimental results show that the PID controller with the given gains ($K_p = 2.000$, $K_i = 0.000$, $K_d = 0.100$) was able to drive the ball toward the reference position in both tests.

However, the system exhibited poor transient performance, characterized by very large overshoot (above 150% in both cases) and failure to settle within the recorded time window.

The main limitations of the current tuning are: the absence of integral gain, which leaves a residual steady-state error, and an insufficient derivative gain, which is unable to adequately dampen the ball's motion before it significantly overshoots the reference. To improve performance, an increase in the derivative gain could help reduce overshoot, and adding an integral term could eliminate the remaining steady-state error. However, care must be taken, as excessive integral action in a nonlinear system like ball-and-beam can introduce additional oscillation and instability.

Overall, the results highlight the challenge of tuning a PID controller for a physically unstable and nonlinear system. The current controller gains provide a starting point for further iterative tuning, and the data obtained from these two runs will serve as a baseline for evaluating improved controller configurations.

VIII. Conclusion

The PID-controlled ball and beam system was tested to investigate the position control performance. From the experimental results, the controller could move the ball to the desired reference position, and the response had some error and overshoot.

For the tested PID gains of $K_p = 2$, $K_i = 0$, and $K_d = 0.1$, the system showed a high overshoot of about 154–159%. The steady-state error was about 1.81–3.03 cm, while the RMSE was about 3.22–4.56 cm. The settling time was not achieved in the recorded responses. The results show that the system response was not completely stable and there was still a considerable amount of oscillation.

The results also show that the PID parameters need further tuning to improve the system performance. Proportional, integral and derivative gains can be used to reduce

overshoot, steady state error, and oscillation when used in appropriate combinations. In conclusion, the experiment is difficult to control the ball and the beam system and the importance of proper PID tuning to get a faster and more stable response.

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